COMPUTER-GENERATED HOLOGRAPHY

(PHYSICS-BASED HOLOGRAM GENERATION AND PHASE RECONSTRUCTION)

TEAM MEMBERS

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PROJECT OVERVIEW

This project focuses on transforming 2D images into digital holograms using computational techniques and deep learning. By generating phase maps and simulating light propagation, the platform reconstructs images to examine how accurately the original visual information is preserved. Using PyTorch-based learning and optimization, the project explores the factors affecting reconstruction quality and identifies opportunities to improve hologram generation.

PROBLEM STATEMENT

- **Phase Generation**: Convert 2D images into suitable holographic phase distributions for image reconstruction.
- **Computational Methods**: Implement point-cloud calculations, iterative phase retrieval, direct phase optimization, and neural-network-based approaches.
- **Reconstruction Quality**: Investigate how different methods and numerical settings affect the reconstructed images.
- **Comparative Analysis**: Examine differences in phase maps, optimization objectives, reconstructed outputs, and computational behaviour.

MOTIVATION

- **Digital Hologram Generation**: Explore how computational techniques can transform 2D images into holographic representations.
- **Method Exploration**: Understand and investigate physics-based, iterative, optimization-based, and deep learning approaches.
- **Phase Visualization**: Study generated phase maps and reconstructed images to understand the hologram generation process.
- **Reconstruction Analysis**: Examine how different methods and computational settings influence reconstruction quality.

FUNDAMENTALS OF COMPUTER-GENERATED HOLOGRAPHY

1. **Holography is a technique for recording and reconstructing light waves using both intensity and phase information, unlike conventional photography, which mainly records intensity.**

2. **Computer-Generated Holography (CGH) uses computational algorithms and mathematical models of light propagation to generate holograms digitally.**

3. **Since 2D images lack phase information, methods such as iterative phase retrieval, direct phase optimization, CNNs are used to generate phase patterns for image reconstruction.**

complex optical field equation :

$$U(x,y) = A(x,y)e^{i\phi(x,y)}$$

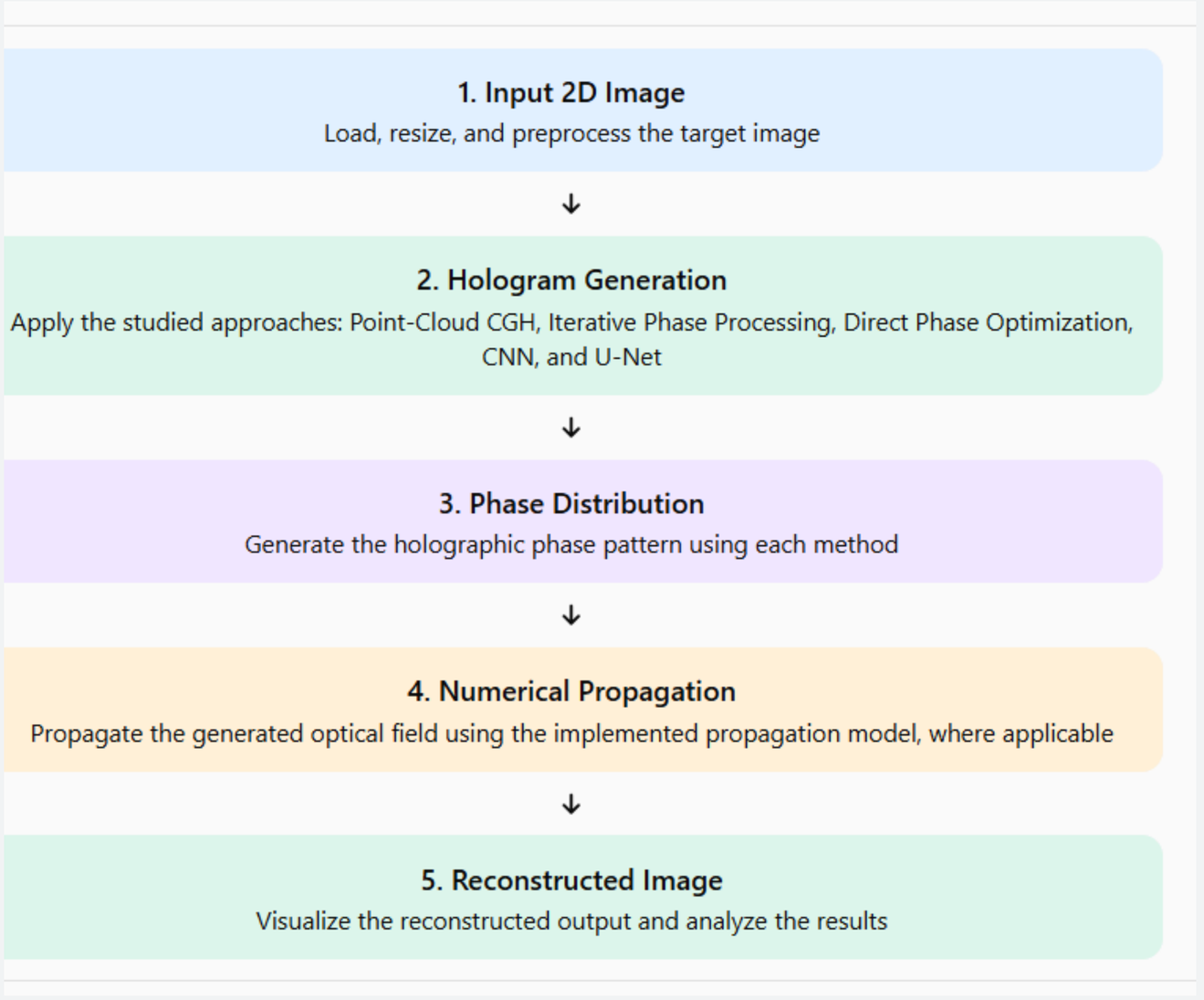
APPROACHES FOR CGH

1. **Point-Cloud CGH**
   Generates a hologram by combining light-wave contributions from discrete object points.

2. **Iterative Phase Processing**
   Repeatedly updates the phase using Fourier-domain operations to obtain the desired reconstruction.

3. **Direct Phase Optimization**
   Optimizes the phase distribution by minimizing the reconstruction error.

4. **CNN-Based Phase Prediction**
   Uses a convolutional neural network to learn phase prediction from image-related inputs.



POINT-CLOUD-BASED HOLOGRAM GENERATION

1. **Input Preparation:** Convert the grayscale image into a (50 * 50) representation and select pixels above a threshold as object points.

2. **Wave Calculation:** Calculate the spherical-wave contribution of each selected point at the hologram plane.

3. **Field Formation:** Sum the complex wave contributions to obtain the total optical field.

4. **Hologram Generation:** Combine the object field with a reference wave to calculate hologram intensity.

Parameter	Value
Wavelength	532 nm
Hologram Size	2 × 2 mm
Pixel Pitch	10 μm
Depth Parameter	-0.2m

ITERATIVE PHASE RETRIEVAL (GERCHBERG–SAXTON ALGORITHM)

1. **Input Image:** Resize the target grayscale image to (128 *128) pixels and use its amplitude information as the reconstruction target.
2. **Phase Initialization:** Assign an initial phase estimate to form a complex optical field.
3. **Forward Transformation:** Apply the Fourier transform to propagate the field into the frequency domain.
4. **Amplitude Constraint:** Replace the transformed field's amplitude with the desired amplitude while retaining its phase.
5. **Inverse Transformation:** Apply the inverse Fourier transform to return to the original domain and enforce the corresponding amplitude constraint.
6. **Iterative Refinement:** Repeat the transformation and constraint-enforcement steps for 50 iterations to refine the phase estimate.

Concept

The Gerchberg–Saxton (GS) algorithm is an iterative phase-retrieval technique that estimates the phase required to reconstruct a desired image by alternating between two domains and enforcing known amplitude constraints.

DIRECT PHASE OPTIMIZATION

1. **Phase Initialization:** Initialize a phase distribution to create the holographic optical field.
2. **Image Reconstruction:** Numerically propagate the field to obtain the reconstructed image.
3. **Loss Calculation:** Calculate the loss between the reconstructed image and the target image
4. **Gradient Computation:** Compute gradients of the loss with respect to the phase values
5. **Phase Optimization:** Use the Adam optimizer to update the phase and reduce the loss iteratively.
6. **Final Output:** Obtain the optimized phase hologram and its reconstructed image.

Concept

Direct phase optimization generates a holographic phase distribution by minimizing the reconstruction error between the target image and the numerically reconstructed image using gradient-based optimization

CNN-BASED PHASE PREDICTION

1. **Input Preparation:** The single-image CNN receives three channels: target amplitude, horizontal coordinate (x), and vertical coordinate (y).

2. **Feature Extraction:** Convolutional layers learn spatial patterns and image features from the input channels.

3. **Phase Prediction:** The network generates a phase map, with the predicted phase bounded using a scaled tanh activation.

4. **Model Training:** The network parameters are updated to reduce the difference between the predicted phase and the optimized target phase using a cosine-based loss function.

5. **Hologram Reconstruction:** The predicted phase is used to form a phase hologram, which can then be evaluated through numerical reconstruction.

Concept

A Convolutional Neural Network (CNN) learns spatial features from input images and uses these learned patterns to predict a holographic phase distribution, which can be used to reconstruct the desired image through numerical light propagation.

SINGLE-IMAGE CNN PHASE PREDICTION

Input Channels
Target Amplitude + X-coordinate + Y-coordinate

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CNN Convolutional Layers
Extract spatial features

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Predicted Phase Map
Bounded using scaled tanh

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Cosine-Based Phase Loss
Compare with optimized phase target

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Update CNN weights and repeat for 5,000 iterations

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Final Phase Prediction
Phase hologram for numerical reconstruction

DATASET-BASED CNN

Image Dataset
2,000 images · 128 × 128

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Training
1,600 images

Validation
400 images

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CNN Model
Convolution + ReLU · 64-channel hidden layers

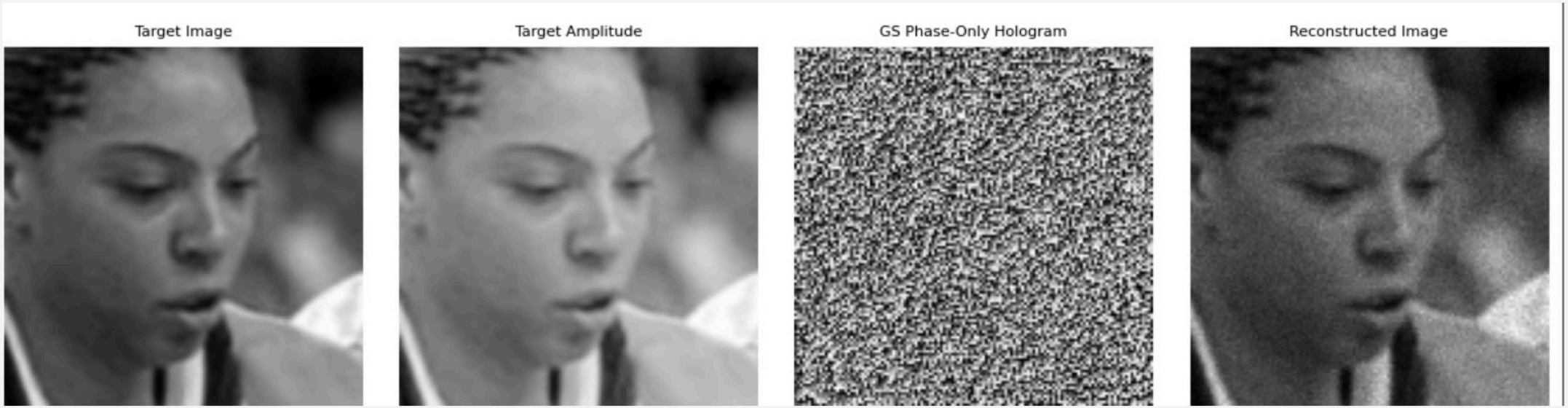
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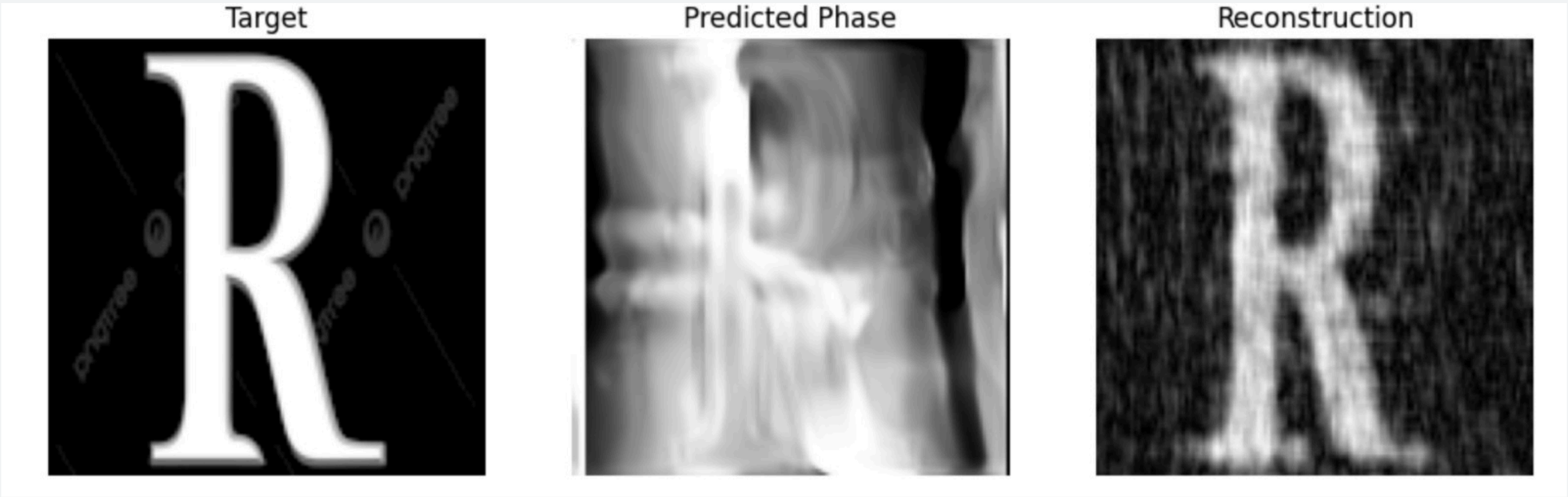
Phase Prediction
Reconstruction-based loss · 5 epochs · Batch size 8

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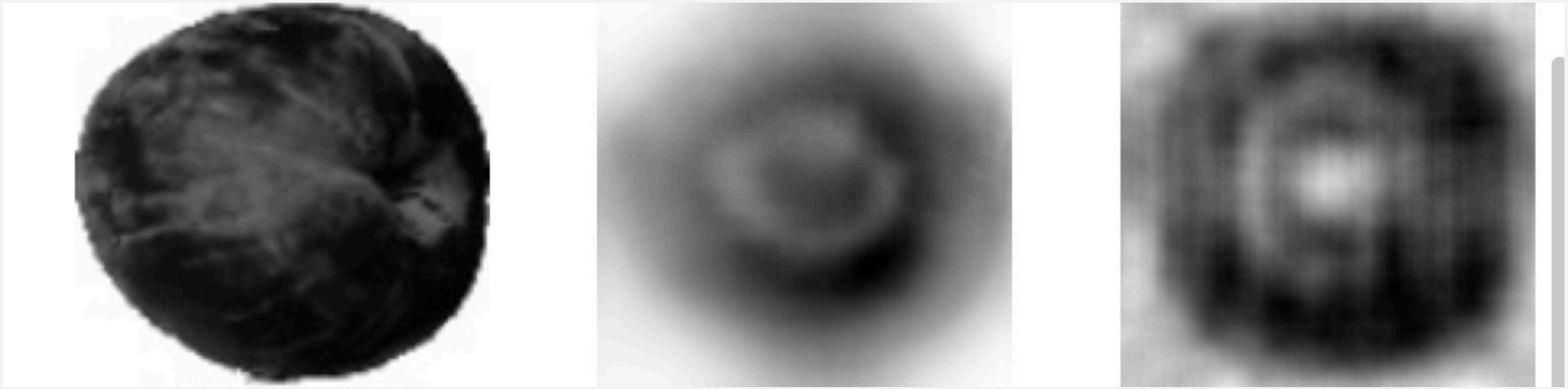
Trained Model
`neural_hologram_model.pth`

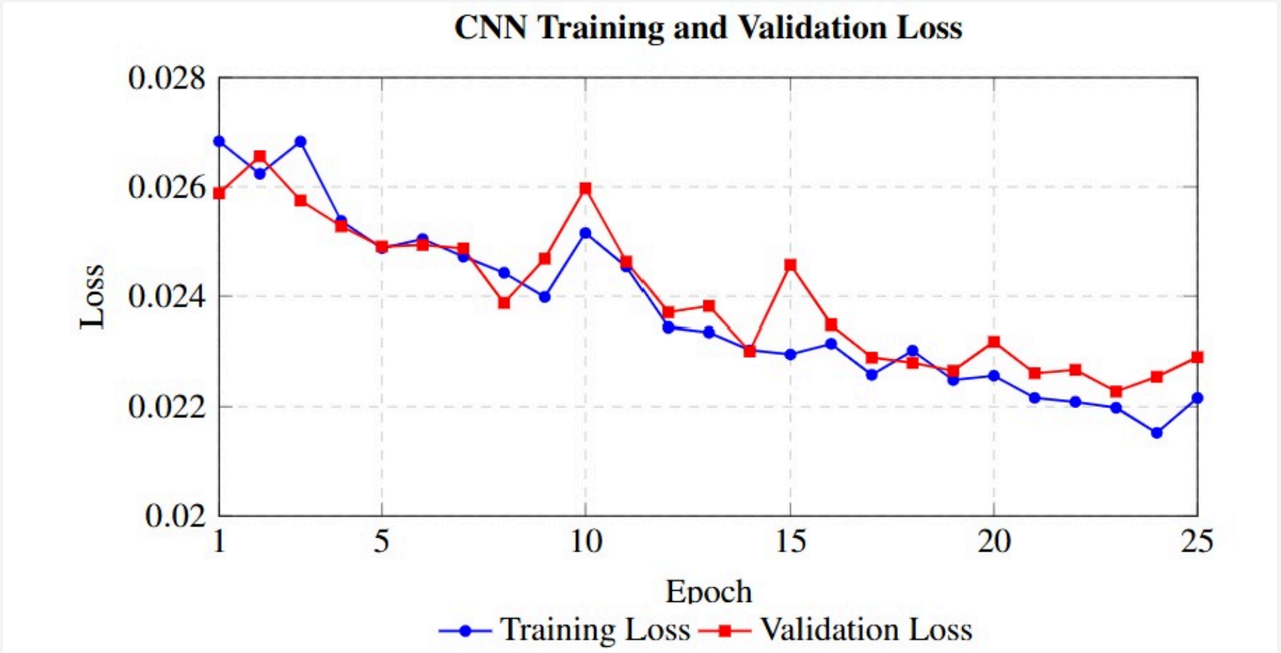
RESULT (GS PHASE VS DIRECT PHASE)





RESULT (CNN BASED)





FUTURE SCOPE

1. **Hologram Encoding and Export:** Convert generated phase maps into device-compatible patterns by wrapping phase values to ([0,2pi]), quantizing them, and exporting them for display on suitable hardware**.**

2. **Physical optical setup and camera in loop optimization**.

3. **Improved CNN Generalization:** Train CNN models on diverse images to improve phase prediction for unseen inputs.

THANK YOU